

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 12 · LECTURE NOTES

Learned Image Processing

What networks changed, and what they invent.

MODULE OUTCOMES

- Explain why learned methods outperform hand-designed ones on these tasks.
- Explain the role of the loss function in what the output looks like.
- Distinguish restoration from synthesis.
- Evaluate learned methods honestly, including their failure modes.
- Explain why perceptual metrics and distortion metrics disagree.

1 Why learned methods work on these problems

EVERY PROBLEM IN THIS COURSE IS UNDERDETERMINED

Demosaicing must produce three colour values per pixel from one measurement (Module 05).

Deconvolution cannot recover frequencies the blur attenuated to zero (Module 06). **Matting** has seven unknowns and three equations per pixel (Module 11). **Monocular depth** is not determined by the image at all.

Every one of them is made solvable by adding a prior — an assumption about what real photographs look like, which selects among the solutions consistent with the measurement.

Hand-designed priors are crude. Smoothness, sparse gradients, piecewise constancy, local colour lines — each captures something real about natural images and each is a drastic simplification of a very complicated distribution.

A network learns the prior from data. What it learns is a far richer description of what photographs look like than anything anyone could write down in closed form. **That is the entire reason learned methods outperform classical ones on these tasks** — not greater cleverness about the forward model, which is well understood, but a better prior over the solutions.

Task	Best classical method	Learned
Denoising	BM3D — genuinely strong and principled, exploiting self-similarity.	Better, particularly at high noise levels where BM3D's patch matching degrades.
Demosaicing	Gradient-corrected interpolation (Module 05).	Noticeably better on difficult texture — exactly where the classical assumption that chrominance varies slowly fails.

Task	Best classical method	Learned
Super-resolution	Interpolation, then sparse coding.	Transformed. The gap is larger here than for any other task on this list — and so is the amount of invention (§3).
Deblurring	Fragile, and heavily dependent on priors (Module 06).	Much better, and still hard.
Matting	Required a user-supplied trimap (Module 11).	Automatic and good , which changed it from a specialist operation into a feature.
The entire ISP	Nine hand-tuned stages (Module 05).	End to end, raw to RGB, in one model.

Joint solving beats staged solving, and this is the strongest technical argument for the end-to-end approach. Demosaicing and denoising performed together outperform either in sequence, because the noise affects the interpolation and the interpolation correlates the noise — so solving one first throws away information the other needed. The classical pipeline's stage boundaries were a decomposition for human comprehensibility, not a property of the problem.

2 The loss function decides the appearance

L2 LOSS PRODUCES BLURRY OUTPUT, NECESSARILY

When several outputs are plausible given the input, the expected squared error is minimised by their average. This is elementary and it has a severe consequence.

The average of several plausible textures is a blur. If a degraded region could plausibly be grass, or gravel, or fabric, the L2-optimal prediction is a smooth grey-green haze that is none of them — because hedging minimises expected error.

So the softness of L2-trained models is not a training failure or a capacity limitation. It is the correct answer to the question that was asked. Asking for minimum expected squared error and receiving a blur is the system working.

Adversarial and perceptual losses ask a different question: produce something that looks like a real photograph, rather than something that minimises expected error. **The output is sharper and less faithful**, and both halves of that are consequences of the change in objective.

Loss	What it optimises	Resulting appearance
L2 / MSE	Expected squared error.	Blurry, and it maximises PSNR — which is why PSNR rankings favour blurry results.
L1	Expected absolute error.	Slightly sharper than L2, because the median is less sensitive to outlying possibilities than the mean.
Perceptual (VGG / LPIPS)	Distance in the feature space of a pretrained network.	Sharper, with plausible texture , because feature-space distance is insensitive to exact pixel placement.

Loss	What it optimises	Resulting appearance
Adversarial (GAN)	Indistinguishability from real images, judged by a discriminator.	Sharpest, and most willing to invent — the objective rewards plausibility, with no term for correspondence to the input.
Diffusion	The full conditional distribution.	Samples from it — so running it twice gives two different plausible answers, which is an honest representation of the underdetermination and an awkward property for a tool.

THE PERCEPTION-DISTORTION TRADEOFF IS A THEOREM

Blau and Michaeli proved that improving perceptual quality necessarily worsens distortion, beyond a certain bound. It is a mathematical result about estimators, not an observation about current architectures.

So 'sharper and more accurate' is not available. Past the bound, a method that looks better than another has measurably higher error against the ground truth, and this cannot be engineered away.

Which explains why PSNR rankings and human visual rankings disagree, sometimes dramatically. They are not measuring the same thing badly; they are measuring genuinely opposed objectives, and a method can only move along the tradeoff curve rather than off it.

And it is why serious papers report both, and why you should always ask which objective a claim is about. 'State of the art' means nothing without saying on which axis.

3 Restoration and synthesis

	Restoration	Synthesis
Recovers	Information that is present but degraded — buried in noise, attenuated by blur, subsampled.	Information that was never measured at all.
Examples	Denoising; mild deblurring; demosaicing; modest upscaling.	Large-factor upscaling; inpainting; recovering frequencies beyond the lens cutoff; filling occluded regions.
The answer is determined by	The measurement , with the prior resolving residual ambiguity.	The prior , with the measurement providing loose constraints.
More training data	Converges toward the true answer.	Changes what is invented, not whether something is invented.

Most real methods do both, often within a single output and without marking the boundary. **That boundary is where the honesty is required**, and it is almost never reported.

THE UPSCALED FACE IS NOT THAT PERSON'S FACE

Take a 16×16 pixel face and upscale it by a factor of eight with a modern model. The result is a sharp, detailed, entirely plausible face.

It is not the face of the person who was photographed. The information distinguishing one face from another at that scale was never recorded — it is below the sampling limit. The model has produced *a* face consistent with the low-resolution evidence and with its training distribution.

Different models produce different faces from the same input, and the same model with a different random seed can too. **That is the proof:** if the information were present in the input, they would agree.

This matters in forensics, in medical imaging, and in journalism, where an image is used as evidence about what was there. A plausible reconstruction presented as a photograph is a claim that has not been earned. **Module 13 takes this up as the subject rather than the caveat.**

- **Distribution shift.** A model trained on one camera's noise characteristics, or one population of scenes, degrades on another — and degrades silently.
- **Confident wrongness.** Most deployed models produce no uncertainty estimate, and **the output looks exactly as plausible when it is wrong as when it is right**. There is no signal to the user that this particular output should not be trusted. **Operationally this is the most serious of the failures**, because every other one could be managed if it announced itself.
- **Hallucinated structure.** Text rendered as text-like marks that are not words, regular patterns found in noise, faces constructed from ambiguous evidence. The model produces what its training distribution says is likely.
- **Averaging away rare content.** Genuinely unusual but real detail is improbable under the learned prior, so it is treated as noise and removed — which is exactly backwards for scientific or forensic imaging, where the unusual thing is the point.

- **And bias from the training distribution**, which for face-related tasks has been demonstrated repeatedly: upscaling and restoration models have been shown to shift facial features toward the demographics over-represented in their training data. **This is not a hypothetical failure** and it is a direct consequence of synthesis from a prior.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Blau & Michaeli — The Perception-Distortion Tradeoff (free)	The theorem in §2. Read it before making any claim that one method is better than another.
Zhang et al. — The Unreasonable Effectiveness of Deep Features as a Perceptual Metric (LPIPS, free)	Perceptual metrics, and a clear demonstration of where PSNR fails to track human judgement.
Chen et al. — Learning to See in the Dark (free)	End-to-end raw-to-RGB in extreme low light. The clearest demonstration of the §1 argument, with a dataset.
Menon et al. — PULSE, and the subsequent discussion (free)	The upscaling case of §3 , including the bias findings that followed publication. Read the paper and the critique together.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Train or download a denoiser with an L2 loss and one with an adversarial loss. Compare PSNR and compare appearance, and confirm the ranking inverts.
- 2 Demonstrate the L2 blurring argument directly: construct a case with two equally plausible outputs and show that the L2-optimal prediction is neither.
- 3 Measure PSNR and LPIPS for several methods and plot them against each other. The tradeoff curve should be visible.
- 4 **Upscale the same low-resolution face with three different models** and compare the results. Document that they disagree.
- 5 Run the same diffusion-based restoration twice with different seeds and compare.
- 6 Apply a denoiser trained on one camera's noise to another camera's raw files and document the degradation.
- 7 Find a case where a learned method removes real but unusual detail. Astronomical or microscopy images are the easiest place to look.

- 8 Compare a learned demosaicer against the Malvar method (Module 05) on difficult texture, and identify where the learned one invents.
- 9 For one output of a learned pipeline, mark the regions that are restoration and the regions that are synthesis, and justify the boundary.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why are learned methods well suited to the problems in this course?
- 2 What is the argument for end-to-end over staged processing?
- 3 Why does L2 loss necessarily produce blur, and why is that not a bug?
- 4 Name five losses and the appearance each produces.
- 5 State the perception–distortion tradeoff and say what kind of result it is.
- 6 Distinguish restoration from synthesis on four axes.
- 7 Why is an upscaled face not that person's face, and what is the proof?
- 8 Name five failure modes of learned methods and say which is most serious operationally.